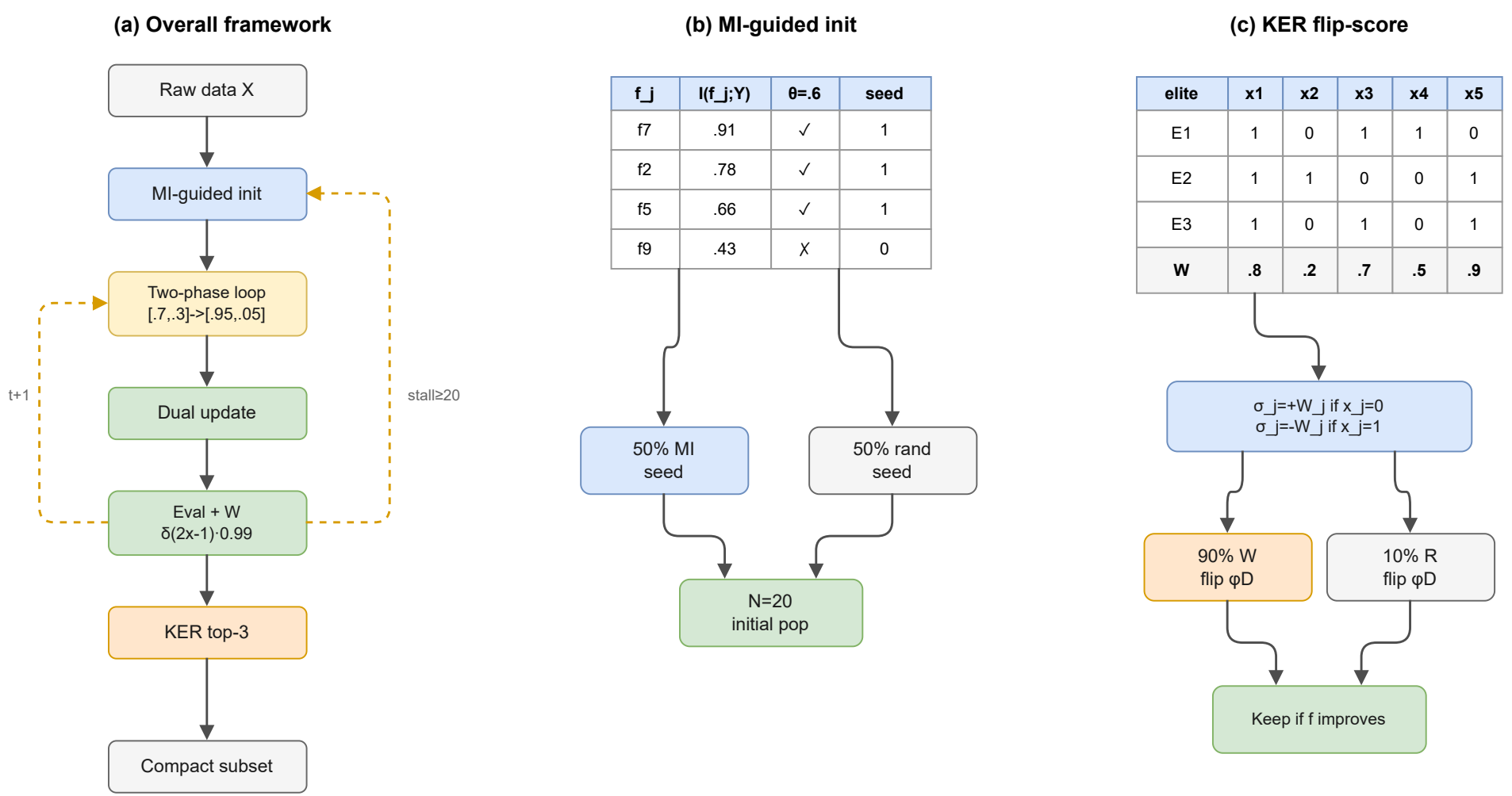

(a) Overall framework
Raw data X
MI-guided init
Two-phase loop
[.7,.3]->[.95,.05]
Dual update
Eval + W
δ(2x-1)·0.99
KER top-3
Compact subset
t+1
stall≥20
(b) MI-guided init
f_j
I(f_j;Y)
θ=.6
seed
f7
.91
✓
1
f2
.78
✓
1
f5
.66
✓
1
f9
.43
✗
0
50% MI
seed
50% rand
seed
N=20
initial pop
(c) KER flip-score
elite
x1
x2
x3
x4
x5
E1
1
0
1
1
0
E2
1
1
0
0
1
E3
1
0
1
0
1
W
.8
.2
.7
.5
.9
σ_j=+W_j if x_j=0
σ_j=-W_j if x_j=1
90% W
flip φD
10% R
flip φD
Keep if f improves